Scalability & Future Plan

Date	29 September 2026
Team ID	SWTID-2026-3161
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Access rules are written as separate ACL scripts per table and field, so every new field value or role needs a script change.	Admin and developer effort grows with each new rule, and rules can become inconsistent.	Medium
2	The ACL script runs on every record read, with no caching of the field-value check.	List views and reports slow down on tables with a very large number of records.	High
3	Denied-access attempts are not logged in detail, and there is no dashboard for reviewing them.	Hard to audit who tried to open restricted records, which weakens compliance checks.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Works well for a small team; the ACL script is evaluated for each user request.	Use role- and group-based rules instead of per-user checks, and test with load scripts so more users can be added without slowdown.
2	Data Storage	Access depends on one field value on each record; data volume is small.	Index the field used in the ACL condition, and archive old records so restricted tables stay lean as data grows.
3	Performance	Script runs on every record read, with no optimization.	Keep the script lightweight, move the condition into the query filter where possible, and cache repeated role lookups.

4	Security	Records are restricted by field value; basic role checks are in place.	Add access logging for denied attempts, periodic ACL reviews, least-privilege roles, and alerts for suspicious access patterns.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Central rule table where admins define field value and role mappings without editing scripts.	1–2 months	Faster rule changes and fewer scripting errors.
Phase 3	Access audit log with a dashboard showing allowed and denied attempts.	3–4 months	Better traceability and compliance reporting.
Phase 4	Extend the same ACL model to other tables and apps, with performance tuning and automated tests.	5–6 months	Consistent security across the platform at larger scale.